

OpenMP Performance

Victor Eijkhout

2026 PCSE

Improvement

- Does your code improve over the sequential?
- Note: The one-thread code need not behave the same as the sequential because of compiler options.
- Can your compiler generate reports? Intel: `-qopt-report`

Scaling

- Run large enough problems; is there overhead in small problems?
- Run for multiple core counts. Do you get linear speedup? Linear up to a certain core count?
- Hint: set `OMP_PROC_BIND=true`, this may help at large core counts.

Testing setup

- The first measurement is likely anomalous: setup, allocation, thread team creation, et cetera.
- First a dummy execution, then a timed one – in the same program run of course.
- (Technically: page instantiation, thread team creation, 'warm cache')